

Norway

SLEIPNER

Volve F

F-9

F-9 A

Plan: F-9A Utsira Water injector Rev1.0 RM KO426m 120609

Standard Survey Report

11 april, 2018

Statoil

Survey Report

Company:	Norway	Local Co-ordinate Reference:	Site Volve F
Project:	SLEIPNER	TVD Reference:	Rotary Table @ 54.90m (Rotary Table)
Site:	Volve F	MD Reference:	Rotary Table @ 54.90m (Rotary Table)
Well:	F-9	North Reference:	Grid
Wellbore:	F-9 A	Survey Calculation Method:	Minimum Curvature
Design:	F-9A Utsira Water injector Rev1.0 RM KO426m 120609	Database:	Production EDM P246N

Project	SLEIPNER, Norway		
Map System:	Universal Transverse Mercator	System Datum:	Mean Sea Level
Geo Datum:	European 1950 - Mean		Using Well Reference Point
Map Zone:	Zone 31N (0 E to 6 E)		Using geodetic scale factor

Site	Volve F, 15/9			
Site Position:		Northing:	6,478,563.53 m	Latitude: 58° 26' 29.8068 N
From: Map		Easting:	435,050.02 m	Longitude: 1° 53' 14.9285 E
Position Uncertainty:	0.00 m	Slot Radius:	13.200 in	Grid Convergence: -0.95 °

Well	F-9			
Well Position	+N/-S	-1.96 m	Northing:	6,478,561.57 m
	+E/-W	1.11 m	Easting:	435,051.13 m
Position Uncertainty	0.00 m		Wellhead Depth:	91.00 m
			Water Depth:	91.00 m

Wellbore	F-9 A				
Magnetics	Model Name	Sample Date	Declination (°)	Dip Angle (°)	Field Strength (nT)
	BGGM2009	27.06.2009	-1.99	71.60	50,401

Design	F-9A Utsira Water injector Rev1.0 RM KO426m 120609			
Audit Notes:				
Version:	Phase:	PLAN	Tie On Depth:	420.40
Vertical Section:	Depth From (TVD) (m)	+N/-S (m)	+E/-W (m)	Direction (°)
	145.90	-1.96	1.11	190.04

Survey Tool Program	Date	01.07.2010		
From (m)	To (m)	Survey (Wellbore)	Tool Name	Description
218.50	420.40	F9 17 1/2in Surveys (F-9)	Keeper, stat	Gyro Tool from SDC
420.40	620.00	F-9A Utsira Water injector Rev1.0 RM KO4	Keeper, stat	Gyro Tool from SDC
620.00	970.70	F-9A Utsira Water injector Rev1.0 RM KO4	Magnetic, std, mag-corr	Magnetic Tools (MWD, EMS)
970.70	1,206.00	F-9A Utsira Water injector Rev1.0 RM KO4	Magnetic, std, mag-corr	Magnetic Tools (MWD, EMS)

Planned Survey										
Measured Depth (m)	Inclination (°)	Azimuth (°)	Vertical Depth (m)	+N/-S (m)	+E/-W (m)	Vertical Section (m)	Dogleg Rate (°/30m)	Build Rate (°/30m)	Turn Rate (°/30m)	
420.40	0.35	352.69	420.40	-1.74	0.83	-0.16	0.000	0.000	0.000	
426.40	0.66	189.63	426.40	-1.76	0.82	-0.15	5.000	1.550	-815.279	
970.60	56.00	190.00	887.01	-246.31	-42.28	248.18	3.051	3.051	0.020	
1,206.00	56.00	190.00	1,018.65	-438.50	-76.17	443.34	0.000	0.000	0.000	

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Well:	F-9	North Reference:	Grid
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Design Targets										
Target Name										
- hit/miss target	Dip Angle	Dip Dir.	TVD	+N/-S	+E/-W	Northing	Easting	Latitude	Longitude	
- Shape	(°)	(°)	(m)	(m)	(m)	(m)	(m)			
F-9A Geological Target 2	1.00	1.00	887.00	-246.18	-42.29	6,478,317.43	435,007.75	58° 26' 21.8291 N	1° 53' 12.5737 E	
- plan misses target center by 0.08m at 970.52m MD (886.97 TVD, -246.25 N, -42.27 E)										
- Polygon										
Point 1			882.80	245.12	7.57	6,478,077.47	434,957.75			
Point 2			887.33	-11.02	52.73	6,478,337.43	434,957.75			
Point 3			887.36	-28.39	-45.78	6,478,337.43	435,057.75			
Point 4			882.83	227.75	-90.95	6,478,077.47	435,057.75			
F-9A Drillers Target 2306	1.00	1.00	887.00	-246.18	-42.29	6,478,317.43	435,007.75	58° 26' 21.8291 N	1° 53' 12.5737 E	
- plan misses target center by 0.08m at 970.52m MD (886.97 TVD, -246.25 N, -42.27 E)										
- Polygon										
Point 1			883.01	230.86	-2.10	6,478,089.83	434,969.74			
Point 2			887.12	-0.92	38.77	6,478,325.06	434,969.74			
Point 3			887.14	-14.13	-36.12	6,478,325.07	435,045.76			
Point 4			883.04	217.65	-76.99	6,478,089.83	435,045.76			

Casing Points						
	Measured Depth	Vertical Depth			Casing Diameter	Hole Diameter
	(m)	(m)	Name		(in)	(in)
	220.80	220.80	30"		30.000	36.000
	421.00	421.00	13 3/8"		13.375	17.500
	991.00	898.42	9 5/8"		9.625	12.250

Formations						
	Measured Depth	Vertical Depth			Dip	Dip Direction
	(m)	(m)	Name	Lithology	(°)	(°)
	970.58	887.00	Utsira		0.00	

Checked By: _____	Approved By: _____	Date: _____
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